

AI Integrated Solar Performance Optimization

ABSTRACT

Solar energy has emerged as a key solution for addressing climate change due to its clean, abundant, and sustainable nature. However, maintaining optimal performance of photovoltaic (PV) systems remains challenging due to environmental variations, system degradation, and unexpected faults. This work presents an integrated artificial intelligence–driven framework for solar PV monitoring and performance optimization, combining anomaly detection, predictive maintenance, and real-time data visualization. Isolation Forest is employed for baseline anomaly detection, while a Long Short-Term Memory (LSTM) model is used to capture temporal patterns in PV system behaviour. A hybrid approach integrating Isolation Forest and LSTM is further developed, resulting in improved anomaly detection performance compared to individual models. To address the limitations of the initial dataset for predictive tasks, an additional dataset is incorporated to implement predictive maintenance and weather forecasting. A Random Forest classifier is used for predictive maintenance to identify potential fault conditions, while a Random Forest regressor is applied for weather forecasting to estimate environmental parameters influencing PV performance. Experimental results demonstrate that the hybrid anomaly detection model achieves superior accuracy and recall, while the predictive models provide reliable insights for maintenance planning and environmental forecasting. The proposed system offers a comprehensive and practical solution for intelligent solar PV system monitoring.

REFERENCES

- [1] A. Syamsuddin, S. M. Budi, and R. Fitriani, "Predictive maintenance based on anomaly detection in photovoltaic system using SCADA data and machine learning," *Energy Reports*, vol. 12, pp. 456–468, Mar. 2024.
- [2] A. Abdessadak, M. El Bouardi, and N. Benhaddou, "Anomaly detection using K-Means and long short-term memory for predictive maintenance of large-scale solar photovoltaic plants," *Energy Reports*, vol. 9, pp. 1298–1312, Jan. 2023.
- [3] M. Castilla, F. Moreno, and A. López, "Artificial neural network-based performance prediction of flat-plate solar collector fields," *Renewable Energy*, vol. 212, pp. 987–996, Apr. 2024.
- [4] M. De Benedetti, F. Leonardi, and S. Santini, "Anomaly detection and predictive maintenance for photovoltaic systems," *Neurocomputing*, vol. 311, pp. 89–101, Sept. 2018

Presented By,

Akshay Kumar P(07)

Manas MS(22)

Nanda V S(24)

Srevya S Deep(36)

Sreyas Sreekumar(44)

Guided By

Mr. Ayush Vijayan

EED, MBCET

Co-Guided By,

Ms. Neetha Chandran

EED, MBCET.